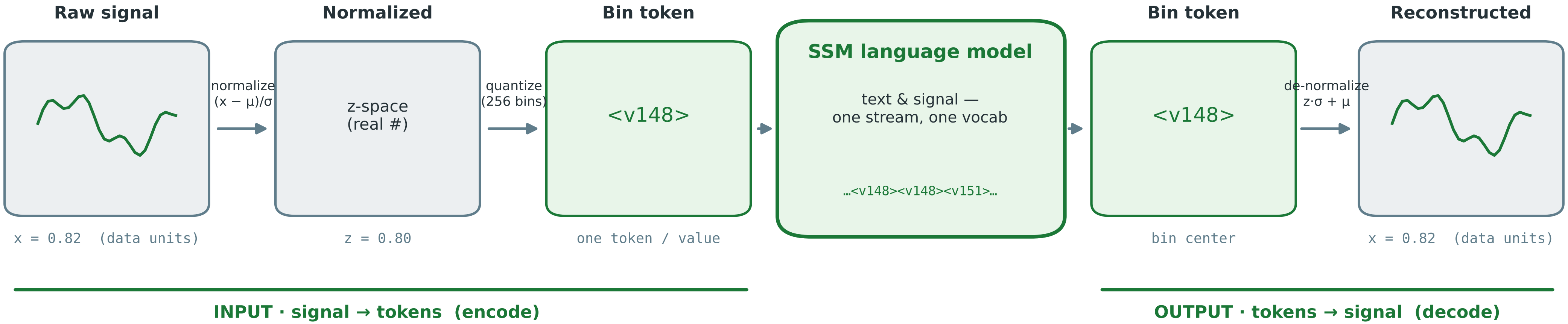
Input & output modality: signal ⇄ z-score ⇄ bin tokens



Measured (A100, 370m unless noted): the stream stays LINEAR at signal scale

Scaling (training step, bs1)

1.2k tok → 0.32 s
6k tok → 0.92 s
12k tok → 1.63 s
≈ 0.12 ms / token

Full 12-lead ECG (12,248 tok)

fits w/o checkpointing
21.5 GB · 1.64 s/step
1.4b: 15.7 GB (ckpt)
throughput RISES w/ length

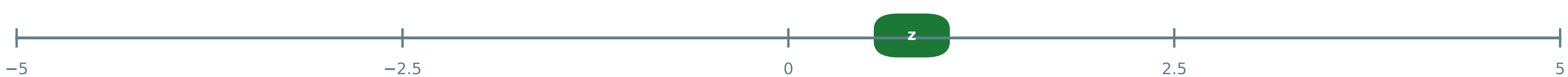
Understanding (TSQA, v4)

Mamba-1.4b 0.9983
SP (paper-faithful) 0.9946
3.2× error reduction
full 4,800-sample test

Generation (held-out forecast)

same vocab, no head
median-of-20 @ T=0.5
−25% MSE vs greedy
(0.48 → 0.36 KS-MSE)

How "quantize" works: z mapped to 1 of 256 uniform bins over z ∈ [−5, 5] · bin embeddings initialized from the LM's own number-text embeddings ("numeracy init")



z = 0.80 → bin 148 / 256 → <v148>

Same 256-token alphabet <v0> … <v255> for input AND output — one representation, no encoder / decoder.   μ, σ shared by encode & decode (forecasts reuse the history's stats).